

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1 Product identifier

Product name

n-Methyl Pyrrolidone (NMP)

Name: N-methyl-2-pyrrolidone (CAS: 872-50-4, EC: 212-828-1, Index: 606-021-00-7)**REACH Registration number:** 01-2119472430-46**Product code**

[100045]

<https://my.chemius.net/p/nf1Z1l/en/pd/en>

1.2 Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses

Solvent.

Uses advised against

No information.

1.3 Details of the supplier of the safety data sheet

Supplier

KH Chemicals B.V.
Scheepmakerij 260
3331 MB Zwijndrecht, Netherlands
+ 31 85 6 255 255
msds@khchemicals.com

1.4 Emergency Telephone Number

Emergency

112

Supplier

+31 85 6255222 (ChemTel - for professionals only)

SECTION 2: HAZARDS IDENTIFICATION

2.1 Classification of the substance or mixture

Classification according to Regulation (EC) No 1272/2008 (CLP)

Skin Irrit. 2; H315 Causes skin irritation.

Eye Irrit. 2; H319 Causes serious eye irritation.

STOT SE 3; H335 May cause respiratory irritation.

Repr. 1B; H360D May damage the unborn child.

2.2 Label elements

Labelling according to Regulation (EC) No 1272/2008 (CLP)**Signal word:** DANGER**Hazard statements:**

H315 Causes skin irritation.
H319 Causes serious eye irritation.
H335 May cause respiratory irritation.
H360D May damage the unborn child.

Supplemental hazard information (EU):

Not applicable.

Precautionary statements:

P261 Avoid breathing mist/vapours/spray.
P280 Wear protective gloves/protective clothing/eye protection/face protection.
P302 + P352 IF ON SKIN: Wash with plenty of soap and water.
P304 + P340 IF INHALED: Remove person to fresh air and keep comfortable for breathing.
P305 + P351 + P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P308 + P313 IF exposed or concerned: Get medical advice/attention.
P405 Store locked up.

Contains:

N-methyl-2-pyrrolidone

Special provisions

Restricted to professional users.

2.3 Other hazards**PBT/vPvB**

No information.

Endocrine disrupting properties

The product does not contain substances with the potential for endocrine disorders.

Additional information

No information.

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS**3.1 Substances**

Name	CAS EC Index REACH	%	Classification according to Regulation (EC) No 1272/2008 (CLP)	Specific Concentration Limits	Notes for substances
N-methyl-2-pyrrolidone	872-50-4 212-828-1 606-021-00-7 01-2119472430-46	100	Skin Irrit. 2; H315 Eye Irrit. 2; H319 STOT SE 3; H335 Repr. 1B; H360D	STOT SE 3; H335; C ≥ 10%	SVHC

3.2 Mixtures

For substances see 3.1.

Notes for substances

SVHC	substance of very high concern
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SECTION 4: FIRST AID MEASURES

4.1 Description of first aid measures

General notes

Take off all contaminated clothing immediately. Never give anything by mouth to an unconscious person. Place patient in recovery position and ensure airway patency.

Following inhalation

Remove patient to fresh air - move out of dangerous area. If victim is not breathing, give artificial respiration. If symptoms develop and persist, seek medical attention. In case of unconsciousness bring patient into stable side position and seek medical attention.

Following skin contact

Immediately remove contaminated clothing. Wash affected skin areas immediately with plenty of water and soap. If symptoms develop and persist, seek medical attention.

Following eye contact

Immediately flush eyes with running water, keeping eyelids apart. If irritation does not stop, seek professional medical treatment!

Following ingestion

Rinse mouth and drink plenty of water (only if the person is conscious). In case of doubt or if feeling unwell seek medical help. Show the physician the safety data sheet or label.

4.2 Most important symptoms and effects, both acute and delayed

Following inhalation

May cause irritation of respiratory system. Coughing, sneezing, nasal discharge, labored breathing.

Following skin contact

Itching, redness, pain.

Following eye contact

Redness, tearing, pain.

Following ingestion

Irritates mucous membranes in the mouth, throat, esophagus and in gastrointestinal area. May cause nausea/vomiting and diarrhea. May cause abdominal discomfort.

4.3 Indication of any immediate medical attention and special treatment needed

No information.

SECTION 5: FIREFIGHTING MEASURES

5.1 Extinguishing media

Suitable extinguishing media

Carbon dioxide (CO₂).

Fire extinguishing powder.

Water spray. Extinguish large fires with water spray or alcohol-resistant foam.

Unsuitable extinguishing media

Full water jet.

5.2 Special hazards arising from the substance or mixture

Hazardous combustion products

In case of a fire toxic gases can be generated; do not inhale gases/smoke. In the event of fire the following can be generated: carbon monoxide (CO), carbon dioxide (CO₂).

Nitrogen oxides (NO_x).

5.3 Advice for firefighters

Protective actions

In case of fire or heating do not breathe fumes/vapours. Vapours can form explosive mixtures with air. Cool containers at risk with water spray. If possible remove containers from endangered area.

Special protective equipment for fire-fighters

Firefighters should wear appropriate protective clothing for firefighters (including helmets, protective boots and gloves) (EN 469) and self-contained breathing apparatus (SCBA) with a full face-piece (EN 137).

Additional information

No information.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1 Personal precautions, protective equipment and emergency procedures

For non-emergency personnel

Protective equipment

Use personal protective equipment (Section 8).

Precautionary measures

Ensure adequate ventilation. Keep away from sources of ignition and/or heat; No smoking!

Emergency procedures

Prevent access to unprotected personnel. Avoid contact with skin and eyes. Do not breathe vapour or mist.

For emergency responders

Use personal protective equipment.

6.2 Environmental precautions

Do not allow product to reach water/drains/sewage systems or permeable soil. If accidental large entry into water or ground occurs, inform responsible authorities.

6.3 Methods and material for containment and cleaning up

For containment

Stem the spill if this does not pose risks.

For cleaning up

Absorb product (with inert material), collect it in special container and dispose it to a licensed hazardous-waste disposal contractor. Make sure the leakage site is well aired.

Other information

See Section 7: HANDLING AND STORAGE.

6.4 Reference to other sections

See also sections 8 and 13.

SECTION 7: HANDLING AND STORAGE

7.1 Precautions for safe handling

Protective measures

Measures to prevent fire

Ensure adequate ventilation. Vapours and air form explosive mixtures. Ignitable mixtures can form in the empty container. Keep away from sources of ignition - no smoking. Take precautionary measures against static discharges.

Measures to prevent aerosol and dust generation

Avoid formation of aerosols.

Measures to protect the environment

No information.

Other measures

No information.

Advice on general occupational hygiene

Use good personal hygiene practices – wash hands at breaks and when done working with material. Do not eat, drink or smoke while working. Avoid contact with skin and eyes. Do not breathe vapours/mist. Avoid exposure - obtain special instructions before using. Refer to instructions on label and regulations for safety and health at work.

7.2 Conditions for safe storage, including any incompatibilities

Technical measures and storage conditions

Keep in cool and well ventilated area. Keep unauthorized personnel away. Keep away from food, drink and animal feeding stuffs. Consider water legal regulations. Observe the regulations for flammable liquids. Keep away from oxidising substances. Keep away from acids. Keep away from flammable substances. Keep in a locked place.

Packaging materials

No information.

Requirements for storage rooms and vessels

Keep in tightly closed containers. Close opened containers after use. Put the containers upright to prevent from leaking.

Storage temperature

No information.

Storage class

No information.

Further information on storage conditions

No information.

7.3 Specific end use(s)

Recommendations

No information.

Industrial sector specific solutions

No information.

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1 Control parameters

Occupational Exposure limit values

Name	mg/m ³	ml/m ³	Short-term value mg/m ³	Short-term value ml/m ³	Remark	Biological Tolerance Values
n-Methyl-2-pyrrolidone (872-50-4)	40	10	80	20	Sk	/

Information on monitoring procedures

BS EN 14042:2003 Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. BS EN 689:2018 Workplace exposure. Measurement of exposure by inhalation to chemical agents. Strategy for testing compliance with occupational exposure limit values. BS EN 482:2021 Workplace exposure. Procedures for the determination of the concentration of chemical agents. Basic performance requirements.

DNEL/DMEL values

For product

No information.

For components

Name	Type	Exposure route	exp. frequency	Remark	Value
N-methyl-2-pyrrolidone	Worker	dermal	long term systemic effects	/	19.8 mg/kg
N-methyl-2-pyrrolidone	Worker	inhalation	long term systemic effects	/	40 mg/m ³
N-methyl-2-pyrrolidone	Worker	dermal	short term systemic effects	/	208 mg/kg
N-methyl-2-pyrrolidone	Worker	inhalation	short term systemic effects	/	80 mg/m ³

PNEC values**For product**

No information.

For components

Name	Exposure route	Remark	Value
N-methyl-2-pyrrolidone	soil	/	0.138 mg/kg
N-methyl-2-pyrrolidone	fresh water	/	0.25 mg/L
N-methyl-2-pyrrolidone	fresh water sediment	/	0.805 mg/kg
N-methyl-2-pyrrolidone	marine water	/	0.025 mg/L
N-methyl-2-pyrrolidone	marine water sediment	/	0.0805 mg/kg

8.2 Exposure controls**Appropriate engineering control****Substance/mixture related measures to prevent exposure during identified uses**

Use good personal hygiene practices – wash hands at breaks and when done working with material. Avoid contact with eyes and skin. Do not breathe vapours/aerosols. Do not eat, drink or smoke while working. Pregnant women should avoid inhalation and skin contact.

Structural measures to prevent exposure

No information.

Organisational measures to prevent exposure

Remove all contaminated clothes immediately and wash them before reuse.

Technical measures to prevent exposure

Provide good ventilation and local exhaust in areas with increased concentration. Keep away from food, drink and animal feeding stuffs.

Personal protective equipment**Eye and face protection**

Tight fitting protective goggles (EN 16321-1:2022/A1:2025).

Hand protection

Protective gloves (BS EN ISO 374). The glove material must be impermeable and resistant to the product / the substance / the preparation. The exact breakthrough time has to be found out by the manufacturer of the protective gloves and has to be observed. Not suitable: leather gloves and thick gloves. Unsuitable gloves: Natural rubber, Chloroprene rubber, Nitrile rubber, PVC gloves.

Appropriate materials

Material	Thickness	Penetration Time	Remark
Butyl rubber	≥ 0.5 mm	≥ 8 h	EN 374

Skin protection

Cotton protective clothing and shoes that cover the entire foot (BS EN ISO 20345:2022).

Respiratory protection

Not needed under normal use and adequate ventilation. In case of insufficient ventilation wear suitable respiratory protection. Wear suitable protective breathing mask (BS EN 136) with filter A2-P2 (BS EN 14387).

Thermal hazards

No information.

Environmental exposure controls**Substance/mixture related measures to prevent exposure**

No information.

Instruction measures to prevent exposure

No information.

Organisational measures to prevent exposure

No information.

Technical measures to prevent exposure

No information.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1 Information on basic physical and chemical properties

Important health, safety and environmental information

Physical state	liquid
Shape	No information.
Colour	colourless
Odour	amine like
Odour threshold	No information.
Melting/freezing point	-24 °C
Boiling point or initial boiling point and boiling range	204 °C
Flammability	245 °C
Lower and upper explosion limit	1.3 — 9.5 % v/v
Flash point	93 °C
Auto-ignition temperature	No information.
Decomposition temperature	No information.
pH	8.5 — 10 at 20 °C, conc. 100 g/l
Viscosity (dynamic)	1.796 mPas at 25 °C
Solubility (Water)	miscible

Partition coefficient n-octanol/water (log value)	-0.46
Vapour pressure	0.29 hPa
Density	1.026 g/cm ³ at 20 °C
Relative vapour/gas density	No information.
Particle characteristics	No information.

9.2 Other information

Information with regard to physical hazard classes

Explosive properties	Product is not explosive. However, formation of explosive air/ vapour mixtures is possible.
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Other safety characteristics

No information.

SECTION 10: STABILITY AND REACTIVITY

10.1 Reactivity

No information.

10.2 Chemical stability

Product is stable under normal conditions of use, recommended handling and storage conditions. To avoid thermal decomposition do not overheat.

10.3 Possibility of hazardous reactions

Reactions with acids, alkalis and oxidants.

10.4 Conditions to avoid

Do not expose to temperatures exceeding 300°C. Do not expose to temperatures exceeding 300°C.

10.5 Incompatible materials

Strong oxidising agents.

Strong acids. Alkalis.

10.6 Hazardous decomposition products

Under normal use conditions no hazardous decomposition products are expected. In case of fire/explosion vapours/gases that pose a health hazard are released. Carbon dioxide; Carbon monoxide.

Nitrogen oxides.

SECTION 11: TOXICOLOGICAL INFORMATION

11.1 Information on hazard classes as defined in Regulation (EC) No 1272/2008

(a) Acute toxicity**For product**

Exposure route	Type	Species	Time	Value	Method	Remark
oral	LD ₅₀	rat	/	3600 mg/kg	/	/

Exposure route	Type	Species	Time	Value	Method	Remark
dermal	LD ₅₀	rabbit	/	8000 mg/kg	/	/
inhalation	LC ₅₀	rat	4 h	> 5.1 mg/L	/	/

(b) Skin corrosion/irritation

No information.

Additional information

Irritant to skin and mucous membranes.

(c) Serious eye damage/irritation

No information.

Additional information

Causes serious eye irritation.

(d) Respiratory or skin sensitisation

No information.

(e) (Germ cell) mutagenicity

No information.

(f) Carcinogenicity

No information.

(g) Reproductive toxicity

No information.

Summary of evaluation of the CMR properties

May damage the unborn child.

(h) STOT-single exposure

No information.

Additional information

May cause respiratory irritation.

(i) STOT-repeated exposure

No information.

(j) Aspiration hazard

No information.

Symptoms related to the physical, chemical and toxicological characteristics

No information.

Interactive effects

No information.

11.2 Information on other hazards**Endocrine disrupting properties**

The product does not contain substances with the potential for endocrine disorders.

Other information

No information.

SECTION 12: ECOLOGICAL INFORMATION

12.1 Toxicity

Acute (short-term) toxicity

For product

Type	Exposure time	Species	Organism	Method	Remark	Value
IC ₅₀	72 h	algae	/	/	/	> 500 mg/L
LC ₅₀	96 h	fish	<i>Leuciscus idus</i>	/	/	4000 mg/L

Chronic (long-term) toxicity

No information.

12.2 Persistence and degradability

Abiotic degradation, physical- and photo-chemical elimination

For product

Environment	Type / Method	Half Time	Evaluation	Method	Remark
water	/	/	Easily eliminated from water.	/	/

Biodegradation

For product

Type	Rate	Time	Evaluation	Method	Remark
-		/	readily biodegradable	/	/
COD	ca. 1600 g O ₂ /g	/	/	/	/
BOD	< 2 mg O ₂ /g	5 days	/	/	/

12.3 Bioaccumulative potential

Partition coefficient n-octanol/water (log value)

For product

Value	Temperature °C	pH	Concentration	Method
-0.46	/	/	/	/

Bioconcentration factor (BCF)

No information.

Additional information

No bioaccumulation expected.

12.4 Mobility in soil

Known or predicted distribution to environmental compartments

No information.

Surface tension

No information.

Adsorption/Desorption

No information.

12.5 Results of PBT and vPvB assessment

No evaluation.

12.6 Endocrine disrupting properties

The product does not contain substances with the potential for endocrine disorders.

12.7 Other adverse effects

No information.

12.8 Additional information**For product**

Do not allow to reach ground water, water courses or sewage system. Water hazard class 1 (self-assessment): slightly hazardous for water.

SECTION 13: DISPOSAL CONSIDERATIONS**13.1 Waste treatment methods****Product / Packaging disposal****Waste chemical**

Disposal must be made according to official regulations: deliver it to authorised collector/remover/transformer of hazardous waste. Do not dispose together with household garbage. Do not allow product to reach drains/sewage systems. Dispose according to regulations.

Waste codes / waste designations according to LoW

No information.

Packaging

Deliver completely emptied containers to approved waste disposal authorities. Clean packaging with water and possibly with soap. The packaging must be disposed in accordance with local or national regulations.

Waste codes / waste designations according to LoW

No information.

Waste treatment-relevant information

No information.

Sewage disposal-relevant information

No information.

Other disposal recommendations

No information.

SECTION 14: TRANSPORT INFORMATION**14.1 UN number or ID number**

ADR/RID	IMDG	IATA	ADN
Not dangerous according to transport regulations.	Not dangerous according to transport regulations.	Not dangerous according to transport regulations.	Not dangerous according to transport regulations.

14.2 UN proper shipping name

ADR/RID	IMDG	IATA	ADN
Not given/not applicable	Not given/not applicable	Not given/not applicable	Not given/not applicable

14.3 Transport hazard class(es)

ADR/RID	IMDG	IATA	ADN
Not given/not applicable	Not given/not applicable	Not given/not applicable	Not given/not applicable

14.4 Packing group

ADR/RID	IMDG	IATA	ADN
Not given/not applicable	Not given/not applicable	Not given/not applicable	Not given/not applicable

14.5 Environmental hazards

ADR/RID	IMDG	IATA	ADN
NO	NO	NO	NO

14.6 Special precautions for user

ADR/RID	IMDG	IATA	ADN
Limited quantities: Not given/not applicable	Limited quantities: Not given/not applicable		Limited quantities: Not given/not applicable

14.7 Maritime transport in bulk according to IMO instruments

ADR/RID	IMDG	IATA	ADN
/	Not given/not applicable	/	/

SECTION 15: REGULATORY INFORMATION**15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture**

- Regulation (EC) No 1907/2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH) (including last amendment Commission Regulation (EU) 2020/878)
- Regulation (EC) No 1272/2008 on classification, labelling and packaging of substances and mixtures
- EH40/2005 Workplace exposure limits (Fourth Edition 2020)

Information according 2004/42/EC about limitation of emissions of volatile organic compounds (VOC-guideline)

not applicable

Ingredients according to Regulation (EC) No 648/2004 on detergents

No information.

Special instructions

Employment restrictions concerning juveniles must be observed. Observe the regulations on employment and protection against dangerous substances for young people, pregnant women and nursing mothers.

15.2 Chemical Safety Assessment

No Chemical Safety Assessment has been carried out for this substance/mixture by the supplier.

SECTION 16: OTHER INFORMATION

Indication of changes

1.4 Emergency Telephone Number 8.1 Control parameters 8.2 Exposure controls 9.1 Information on basic physical and chemical properties 9.2 Other information 11.1 Information on hazard classes as defined in Regulation (EC) No 1272/2008 12.1 Toxicity 12.2 Persistence and degradability 12.3 Bioaccumulative potential

Key literature references and sources for data

No information.

Abbreviations and acronyms

ATE - Acute Toxicity Estimate
ADR - Agreement concerning the International Carriage of Dangerous Goods by Road
ADN - European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways
CEN - European Committee for Standardisation
C&L - Classification and Labelling
CLP - Classification Labelling Packaging Regulation; Regulation (EC) No 1272/2008
CAS# - Chemical Abstracts Service number
CMR - Carcinogen, Mutagen, or Reproductive Toxicant
CSA - Chemical Safety Assessment
CSR - Chemical Safety Report
DMEL - Derived Minimal Effect Level
DNEL - Derived No Effect Level
DPD - Dangerous Preparations Directive 1999/45/EC
DSD - Dangerous Substances Directive 67/548/EEC
DU - Downstream User
EC - European Community
ECHA - European Chemicals Agency
EC-Number - EINECS and ELINCS Number (see also EINECS and ELINCS)
EEA - European Economic Area (EU + Iceland, Liechtenstein and Norway)
EEC - European Economic Community
EINECS - European Inventory of Existing Commercial Substances
ELINCS - European List of notified Chemical Substances
EN - European Standard
EQS - Environmental Quality Standard
EU - European Union
Euphrac - European Phrase Catalogue
EWC - European Waste Catalogue (replaced by LoW – see below)
GES - Generic Exposure Scenario
GHS - Globally Harmonized System
IATA - International Air Transport Association
ICAO-TI - Technical Instructions for the Safe Transport of Dangerous Goods by Air
IMDG - International Maritime Dangerous Goods
IMSBC - International Maritime Solid Bulk Cargoes
IT - Information Technology
IUCLID - International Uniform Chemical Information Database
IUPAC - International Union for Pure Applied Chemistry
JRC - Joint Research Centre
Kow - octanol-water partition coefficient
LC50 - Lethal Concentration to 50 % of a test population
LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose)
LE - Legal Entity
LoW - List of Wastes (see <http://ec.europa.eu/environment/waste/framework/list.htm>)
LR - Lead Registrant
M/I - Manufacturer / Importer
MS - Member States
MSDS - Material Safety Data Sheet
OC - Operational Conditions
OECD - Organization for Economic Co-operation and Development
OEL - Occupational Exposure Limit
OJ - Official Journal

OR - Only Representative

OSHA - European Agency for Safety and Health at work

PBT - Persistent, Bioaccumulative and Toxic substance

PEC - Predicted Effect Concentration

PNEC(s) - Predicted No Effect Concentration(s)

PPE - Personal Protection Equipment

(Q)SAR - Qualitative Structure Activity Relationship

REACH - Registration, Evaluation, Authorisation and Restriction of Chemicals (Regulation (EC) No 1907/2006)

RID - Regulations concerning the International Carriage of Dangerous Goods by Rail

RIP - REACH Implementation Project

RMM - Risk Management Measure

SCBA - Self-Contained Breathing Apparatus

SDS - Safety data sheet

SIEF - Substance Information Exchange Forum

SME - Small and Medium sized Enterprises

STOT - Specific Target Organ Toxicity

(STOT) RE - Repeated Exposure

(STOT) SE - Single Exposure

SVHC - Substances of Very High Concern

UN - United Nations

vPvB - Very Persistent and Very Bioaccumulative

List of relevant H phrases

H315 Causes skin irritation.

H319 Causes serious eye irritation.

H335 May cause respiratory irritation.

H360D May damage the unborn child.



- ☒ Provided correct labelling of the product
- ☒ Compliance with the local legislation
- ☒ Provided correct classification of the product
- ☒ Provided adequate transport data

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The information of this SDS is based on the present state of our knowledge and meets the requirements of EU and national laws. The user's working conditions however, are beyond our knowledge and control. The product is not to be used for purposes other than those specified under section 1 without a written permission. It remains the responsibility of the user to ensure that the necessary steps are taken to meet the laws and regulations. Handling of the product may only be done by people above 18 years of age, who are satisfactorily informed of how to do the work, the hazardous properties and necessary safety precautions. The information given in this SDS is to describe the product only in terms of health and safety requirements and should not, therefore, be construed as guaranteeing specific properties.